

Problem 61. Find all prime number(s) p such that $p^2 - p - 1$ is the cube of some integer.

Solution. The solutions are $p = 2, 37$.

Let $p^2 - p - 1 = n^3$ for some integer n . Then

$$p(p-1) = n^3 + 1 = (n+1)(n^2 - n + 1). \quad (1)$$

If $p \mid n+1$, then $n^2 - n + 1 \mid p-1$. It follows that $n^2 - n + 1 \leq p-1 \leq n$, i.e. $(n-1)^2 \leq 0$. We must have $n = 1$, and hence $p = 2$. One checks that $(p, n) = (2, 1)$ is a solution.

If $p \mid n^2 - n + 1$, then $n+1 \mid p-1$. Let $p = k(n+1) + 1$ for some positive integer k . Equation (1) becomes

$$(kn + k + 1)k(n+1) = (n+1)(n^2 - n + 1),$$

which can be rewritten as

$$n^2 - (k^2 + 1)n - (k^2 + k - 1) = 0. \quad (2)$$

Since k and n are integers, the discriminant $\Delta = (k^2 + 1)^2 + 4(k^2 + k - 1) = k^4 + 6k^2 + 4k - 3$ is a perfect square. Note that

$$(k^2 + 3)^2 = k^4 + 6k^2 + 9 < k^4 + 6k^2 + 4k - 3 < k^4 + 8k^2 + 16 = (k^2 + 4)^2$$

for $k > 3$. Therefore, it suffices to check $k = 1, 2, 3$. One finds that equation (2) has an integer solution only when $k = 3$. In that case $n = 11$, and $p = 37$. One checks that $(p, n) = (37, 11)$ is a solution. $\square$

Problem 62 (Turkey TST 2019 Problem 8). Let $p > 2$ be a prime number, $m > 1$ and n be positive integers such that $\frac{m^{pn} - 1}{m^n - 1}$ is a prime number. Show that

$$pn \mid (p-1)^n + 1.$$

Solution. Suppose $n = p^a b$ where $a \geq 0$ and $p \nmid b$. First we assume $b > 1$. By Zsigmondy's theorem, there exists a prime q such that $q \mid m^{p^{a+1}} - 1$ but $q \nmid m^{p^a} - 1$. Since

$$(m^{p^{a+1}} - 1, m^n - 1) = m^{(p^{a+1}, n)} - 1 = m^{p^a} - 1,$$

we have $q \nmid m^n - 1$. Then since $m^{p^{a+1}} - 1 \mid m^{p^{a+1}b} - 1 = m^{pn} - 1$, we have $q \mid \frac{m^{pn} - 1}{m^n - 1}$. Also, by Zsigmondy's theorem, there

exists a prime r such that $r \mid m^{pn} - 1$ but $r \nmid m^n - 1$ and $r \nmid m^{p^{a+1}} - 1$, unless $(m, p, n) = (2, 3, 2)$ (rejected since $\frac{2^6 - 1}{2^2 - 1} = 21$ is not a prime). This implies $r \mid \frac{m^{pn} - 1}{m^n - 1}$ and $r \neq q$. Thus, $qr \mid \frac{m^{pn} - 1}{m^n - 1}$, so that $\frac{m^{pn} - 1}{m^n - 1}$ is not a prime.

Therefore, $b = 1$. We now show that $pn = p^{a+1} \mid (p-1)^{p^a} + 1 = (p-1)^n + 1$. In fact, since n is odd, we can apply the lifting the exponent lemma to obtain

$$v_p((p-1)^n + 1) = v_p((p-1) + 1) + v_p(n) = 1 + a,$$

so that $p^{a+1} \mid (p-1)^n + 1$ as desired. $\square$

Problem 63 (Iran TST 2018 Test 2 Problem 3). Let $a_1, a_2, a_3, \dots$ be an infinite sequence of distinct integers. Prove that there are infinitely many primes p that distinct positive integers i, j, k can be found such that $p \mid a_i a_j a_k - 1$.

Solution. We say that an infinite set of integers is good if there are only finitely many primes dividing some of the terms of this set. WLOG assume $a_i \neq 0$ for all i (otherwise remove it). Suppose on the contrary that $\{a_i a_j a_k - 1 : i, j, k \text{ are distinct}\}$ is good. Then $\{a_1 a_2 a_k - 1 : k \geq 4\}$ and $\{a_1 a_3 a_k - 1 : k \geq 4\}$ are both good. Since $a_3 \neq 0$ and $a_2 \neq 0$, the sets $\{a_3(a_1 a_2 a_k - 1) : k \geq 4\}$ and $\{a_2(a_1 a_3 a_k - 1) : k \geq 4\}$ are good. This means $\{a_1 a_2 a_3 a_k - a_3 : k \geq 4\}$ and $\{a_1 a_2 a_3 a_k - a_2 : k \geq 4\}$ are good. This contradicts Kobayashi's theorem since the two sets are just translations of each other. The assertion follows. $\square$

Problem 64 (All-Russian MO 2012 Grade 11 Problem 8). For a positive integer n define $S_n = 1! + 2! + \cdots + n!$. Prove that there exists an integer n such that S_n has a prime divisor greater than 10^{2012} .

Solution. Suppose on the contrary that all prime divisors of S_n are less than 10^{2012} . Let $p_1, p_2, \dots, p_k$ be these prime divisors, which must be odd. WLOG assume $v_{p_i}(S_n)$ is bounded for $i = 1, 2, \dots, t$, and is unbounded for $i = t+1, \dots, k$. Since S_n is unbounded, we must have $t < k$. For simplicity, we write $p = p_i$ for some $t+1 \leq i \leq k$.

For any n , consider the relation $S_n + (n+1)! = S_{n+1}$. If $v_p(S_n) < v_p((n+1)!)$, then $v_p(S_n) = v_p(S_{n+1})$. This implies $v_p(S_{n+1}) < v_p((n+1)!) \leq v_p((n+2)!)$. Inductively, we find that $v_p(S_n) = v_p(S_{n+1}) = v_p(S_{n+2}) = \cdots$. This contradicts the assumption that $v_p(S_n)$ is unbounded. Therefore, we must have

$$v_p(S_n) \geq v_p((n+1)!). \quad (3)$$

Let $m \equiv -2 \pmod{p_1 p_2 \cdots p_k}$ be sufficiently large. Then $p \mid m+2$, so that

$$v_p((m+1)!) < v_p((m+2)!) \leq v_p(S_{m+1}),$$

where the last inequality follows from (3). We now consider the relation $S_m + (m+1)! = S_{m+1}$. From above, we must have $v_p(S_m) = v_p((m+1)!)$. This is equal to $v_p(m!)$ since $p \nmid m+1$.

We have shown that $v_p(S_m) = v_p(m!)$ for $p = p_{t+1}, \dots, p_k$. Also, it is clear that $v_p(S_m) \leq v_p(m!)$ for $p = p_1, \dots, p_t$ and sufficiently large m because the left-hand side is bounded while the right-hand side is not. Therefore, $v_{p_i}(S_m) \leq v_{p_i}(m!)$ holds for each p_i and large $m \equiv -2 \pmod{p_1 p_2 \cdots p_k}$. This implies $S_m \leq m!$, which is a contradiction. Therefore, some S_n has a prime divisor which is greater than 10^{2012} . $\square$